MIPS RF Head Piece Wise Linear Calibration
March 31, 2024

This document describes the procedure to calibrate a RF head monitor values or often called readbacks. The RF head has an internal calibration adjustment that is set in the factory using a capacitor load. Your actual configuration can impact this factory calibration and the following procedure allows you to set the calibration using your actual operating conditions.

You will need to monitor the RF level near the actual load to get an accurate reading, no not use the voltage reading at the test points on the RF heads. These test points are intended for tuning and making sure the waveforms are correct. When using low capacitance loads like quads you can use the test pointes for this calibration because the coupling losses are very low for low capacitance loads.

Calibrate using the following procedure:

1.) Power up the MIPS system and make sure your RF connections are made and secure.
2.) Attached a scope to the RF output channels you wish to calibrate. Make sure your scope probes have been compensated.
3.) Connect the MIPS app to your MIPS box that you are calibrating and select the Terminal tab. Issue to the GVER command to make sure communications is established.
4.) With the scope probes attached use the auto tune procedure to tune the RF head. You can do this using the MIPS front panel control or enter the follow auto tune command.
    TUNERFCH,Channel
    Channel is the RF channel you wish to calibrate.
5.) Enter the following command to start the calibration procedure:
    RFPWLC,Channel,Phase
    Channel is the RF channel you wish to calibrate.
    Phase is the *RF+* or *RF-* depending on the phase you are calibrating. You will need to calibrate both phases.
6.) You will see the following message indicating the system has entered the calibration procedure:
    This function will generate a piecewise linear
    calibration table. Set the drive level to reach
    desired calibration points and then enter the
    measured value to the nearest volt. Press enter
    When finished. Voltage must be increasing!

    Point 1
    Enter measured voltage:
7.) Use the MIPS RF channel drive control to adjust the RF output level for to the desired RF level then enter the measured voltage to the nearest volt peak to peak. You will need a minimum of 3 points and a maximum of ten points. Each point you enter must be greater than the previous point. It is always a good idea to define the first point with 0 drive level, note the RF output will likely be a few volts even with 0 RF drive.

Use calibration points than cover the range you intend to operate. After you enter a point you will see a prompt for the next point. If you wish to terminate the enter just press enter with no input value, you do not need to enter the maximum number of points. After the entry is complete the system will report a summary of the voltages entered and there corresponding ADC readback values.

Note: You cannot use the keyboard backspace key if you make a data entry error, this is a very simple data entry interface so use care when entering the values. If you make an entry error you will need to exit the procedure by pressing enter when prompted for a value and starting again.

8.) You must calibrate both the RF+ and RF- phases, repeat steps 5 through 7 for the RF- phase. If you are using both RF channels on the module you are calibrating then you should calibrate each channel. This is because each RF driver module can drive two RF heads. If your MIPS system has two RF head connections then is has only one module if it has four RF head connections then it has two modules. Make sure and calibrate both RF channels on the module you are working with.

9.) After you have finished the calibration procedure save your changes using the following command:
SAVEM,RF

10.) After you have completed the calibration procedure for both channels on a module you need to set the module firmware revision level for the changes to take effect. Enter the *ABOUT* command and note the board and address values for the RFdriver modules. The first module listed is RF channels 1 and 2 and the second module is RF channel 3 and 4. To set the firmware revision level use the following command:
SMREV,board,address,5
The *board* value will be *a* or *b* that you recorded from the *ABOUT* command. The *address* will likely be *50*, again recorded from the *ABOUT* command report.

11.) This completes the calibration procedure; you should now restart the MIPS system to start using the new calibration data.